VRADICAL STUDIO
USER MANUAL / VERSION 1.0.0

The complete creator guide

VRadical Studio

Camera-native VR180 and flat 2D/3D video from first import to calibrated, stabilized, polished delivery - with photos, mixed-camera Timeline editing, publishing, CLI, and local DeoVR streaming.



16 CHAPTERS	10 REAL CAPTURES	macOS + WINDOWS	GPU-FIRST

Inside this guide

16 practical chapters, written for the person who wants to create rather than decode a technical manual.

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TIP	Use the PDF bookmarks or the HTML search field to jump straight to a camera, tool, or error message.

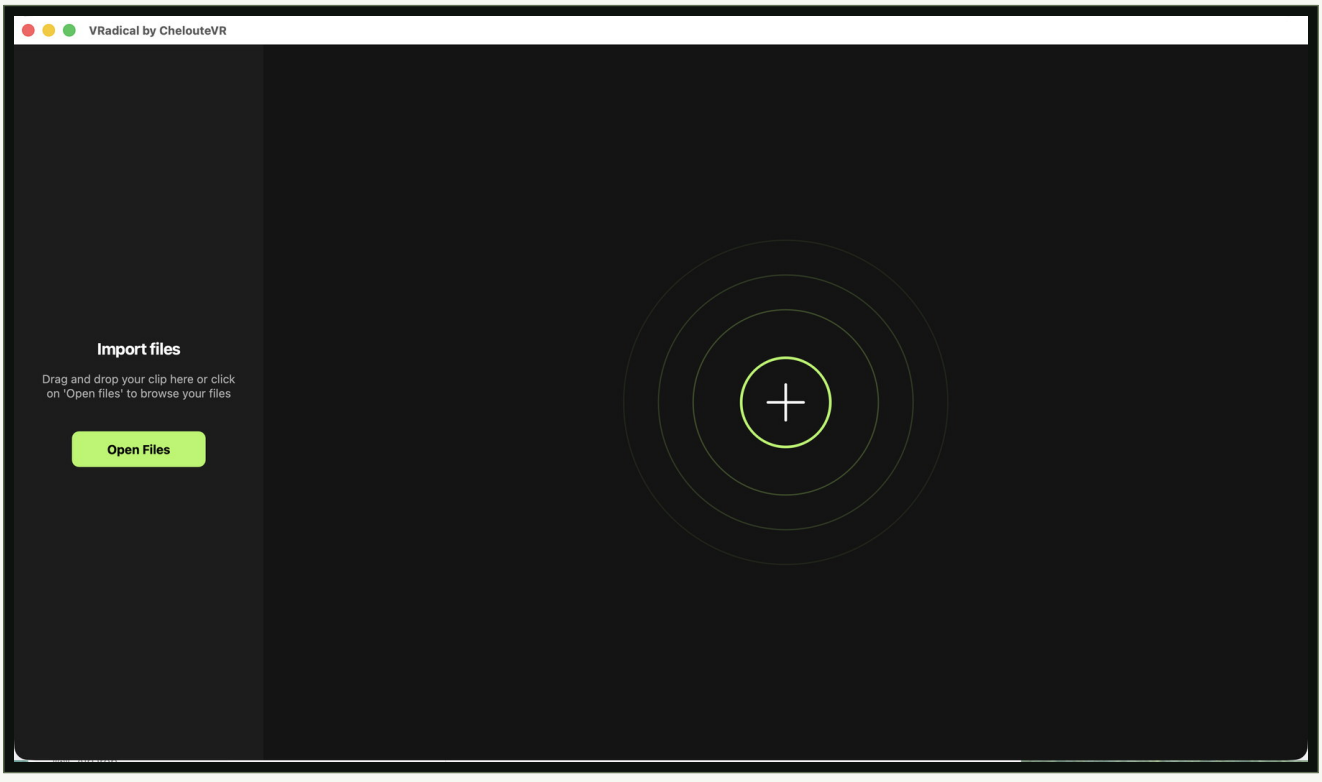
01

FROM CAMERA FILE TO POLISHED VR180

Start here

Welcome to VRadical Studio. This guide is your friendly, practical map of the whole app: importing camera media, calibrating stereo geometry, stabilizing motion, shaping the look, editing a mixed-camera timeline, exporting, and streaming to DeoVR.

Preview Open supported camera originals and inspect the final side-by-side VR180 image on the GPU.	**Polish** Calibrate, stabilize, grade color, correct framing, process audio, denoise, and sharpen.
Build Combine compatible videos, photos, and external audio in a reusable Timeline project.	**Deliver** Export VR180, flat 2D or 3D video, and photos; publish eligible immersive exports or stream locally to DeoVR.



The welcome screen. Drop supported media anywhere in the center, or choose Open Files.

Your first ten minutes

1	**Launch Studio** Open the application and confirm that it reaches the import screen without a GPU warning.
2	**Add media** Drop one or more supported camera files, or choose Open Files. Keep every required companion or segment file beside the selected file.

3	**Complete calibration** If Studio asks for stereo calibration or sensor orientation, finish that guided flow before judging the final picture.
4	**Choose the motion** Open Stabilization, enable VRadical Flow when needed, then add Lock Vertical Direction and/or Chicken Head Stabilization as intentional modifiers.
5	**Shape the image** Open Media Processing for color, LUT, image correction, and camera-specific geometry. Enable RGB histogram when you want to monitor the final left eye.
6	**Set a range** Use the left and right bracket controls to mark In and Out points when you only need part of a video.
7	**Deliver** Choose a VR or flat projection in Export, build a Timeline, or start a DeoVR stream. A higher export frame rate automatically invokes GPU RIFE interpolation.

TIP

The golden rule

Keep camera originals unchanged. Do not rename, trim, transcode, or separate companion files before import. Studio reads camera metadata, stereo tracks, motion data, and multipart identity directly from the originals.

A tiny glossary before we roll

Term	What it means in Studio
Camera pipeline	The named parser, preview, stabilization, calibration, export, Timeline, and streaming route for one camera profile.
SBS / VR180	A side-by-side stereo frame: the final left eye on the left half and final right eye on the right half.
Hequirect	Studio's half-equirectangular VR180 projection. It is the usual final SBS view.
Calibration	A persistent camera-specific stereo correction that aligns parallax and must be applied as the final geometry step.
Mod profile	Saved per-camera setup such as sensor rotations, eye order, and focal-distance defaults.
Timeline	A .vradical-timeline project containing explicit video/photo clips, external audio, trims, transitions, and per-clip processing.
Flat 2D / 3D	Rectilinear video delivery: one 16:9 eye for 2D or two 16:9 eyes side by side for 3D Flat.
RIFE	GPU frame interpolation selected automatically when video export frame rate exceeds the source rate.

02

A GPU IS NOT OPTIONAL HERE

Install and launch

VRadical Studio is GPU-first. Preview, geometry, stabilization, calibration application, color, LUTs, Timeline playback, and display stay on the native GPU path. Unsupported hardware stops at launch instead of silently reducing quality.

Platform	Required hardware	Install location
macOS	Apple Silicon Mac. Intel Macs are not supported.	/Applications/VRadical Studio.app
Windows	Hardware Direct3D 11 GPU. Non-Apple PCs may use NVIDIA, AMD, or Intel; a physical Apple computer running Windows may use its installed hardware GPU.	C:\Program Files\VRadical Studio

> **NOTE**
> **Vision Pro output**
> Spatial and Immersive Vision Pro export is macOS-only and requires Apple Silicon with macOS 14 or later.

Install on macOS

1. **Open the DMG**
   Use the VRadical Studio disk image supplied for your edition.
2. **Copy the app**
   Drag VRadical Studio into Applications. Keep the application bundle intact because it contains the GUI, CLI, libraries, and calibration tools.
3. **Launch**
   Open VRadical Studio from Applications. If macOS asks for confirmation, verify that you are opening the delivery you received.
4. **Upgrade**
   In an updater-enabled customer build, choose Help > Check for Updates or open Settings > General > Updates. A manually delivered build can still be replaced while Studio is closed.

Install on Windows

1. **Run the installer**
   Open VRadicalStudioSetup.exe. Windows may show an unknown-publisher UAC prompt for the current unsigned installer.

2	**Choose the destination** The default is Program Files. The installer can create Start menu and desktop shortcuts.
3	**Finish prerequisites** Allow the installer to add the Microsoft Visual C++ x64 runtime when needed.
4	**Upgrade** Use Help > Check for Updates in an updater-enabled customer build, or quit Studio and run a newer installer over the existing installation.

First-launch checks

- [ ] The import screen appears and remains responsive.
- [ ] No Apple Silicon or Direct3D 11 compatibility message appears.
- [ ] The version shown in Settings matches the delivery notes.
- [ ] Your camera family is enabled in this edition. Camera support is build-specific.
- [ ] There is enough fast local storage for camera originals, exports, and temporary calibration data.

Full version and demo editions

Edition	What you can do
Full version	Use the registered camera and photo routes in this guide, file export and publishing, GPU RIFE interpolation, and DeoVR streaming when the purchased camera family is enabled.
Universal demo	Preview and edit supported camera video and eligible photos, calibrate, use most stabilization and image/audio tools, edit mixed Timelines, and stream to DeoVR. The export settings dialog opens, but submission explains that file export needs the full version. Canon/CALF stabilization and DJI/GoPro photo routes are absent.
Legacy single-camera demo	Only its named camera model's video routes are included. Photo processing, file export, and DeoVR streaming are absent.

WARNING

No CPU preview mode

If the GPU check fails, update the GPU driver or move the job to supported hardware. Only encoding may use a software fallback; image processing never does.

NOTE

Vuze XR requires a Vuze-enabled delivery

Vuze XR is included only when an original Vuze camera file is supplied to prepare the build or release. An ordinary build or universal demo does not include Vuze automatically. If you need this camera, request a Vuze-enabled delivery with an untouched original. Importing a file or changing a setting cannot add a camera that is absent from the installed edition.

03

EXPLICIT SUPPORT, NEVER A GENERIC GUESS

Cameras and media

An extension is only the first gate. Studio also verifies camera metadata, the installed camera family, and complete support for the operation you requested. That strictness keeps preview and export honest.

Accepted containers

Family	Video	Photo	Important notes
GoPro	.360	.36p	MAX2 BERS1, BERS0, and Fisheye are distinct video profiles. Supported single-member recordings have Timeline routes.
DJI Osmo 360 / 360 II	.osv	.jpg (Osmo 360)	Both models have distinct video profiles. The registered JPG photo route is for the original Osmo 360. Keep every OSV member together.
Insta360	.insv	.insp, .dng (X5/X6)	Video profiles include EVO, X4, X5, X6, and One R/RS variants. X5/X6 have photo routes; split video needs matching parts.
Kandao Q3U	.mp4, .mov	.jpg, .dng	MP4/MOV support is metadata-validated. Q3U video support currently targets 24/25/30 fps families.
Lenovo Mirage	.mp4, .mov	.jpg	Video and Extended-XMP photo routes require the matching explicit profile.
Canon EOS VR System	.crm	.jpg, .cr3 (R5 Mark II)	R5 Mark II supports video and photos; R5 C currently supports the registered CRM video mode only. Both require the RF 5.2mm F2.8 L Dual Fisheye.
CALF Gen 1	V#######.mp4	-	Optional camera slice. Availability depends on the purchased edition.
Vuze XR VR180	.mp4	.jpg + .jp2, .dng + .dn2, or assembled .jpg	Vuze-enabled delivery only. Keep paired photos together. Vuze 360 captures are not supported.
Timeline	.vradical-timeline	-	A project references its original media; it is not a generic JSON file.

WARNING

Extensions that are deliberately not aliases

Do not rename .jpg to .jpeg, a Timeline project to .json, or a camera file to a generic MP4 name in the hope of bypassing detection. Shared .jpg, .mp4, and .mov containers are accepted only when their camera metadata matches an enabled pipeline.

Video profile coverage

Profile	Preview	Export	Timeline	DeoVR	Calibration
GoPro MAX2 BERS1 EAC	Yes	Yes	Yes	Yes	Required
GoPro MAX Fisheye	Yes	Yes	Yes	Yes	Required
GoPro BERS0 / older EAC	Yes	Yes	Yes	Yes	Required
DJI Osmo 360 dual-fisheye	Yes	Yes	Yes	Yes	Required
DJI Osmo 360 II dual-fisheye	Yes	Yes	Yes	Yes	Required
Insta360 EVO, X4, X5, X6	Yes	Yes	Yes	Yes	Required
Insta360 One R/RS 1-inch H/V	Yes	Yes	Yes	Yes	Required
Kandao Q3U	Yes	Yes	Yes	Yes	Required
Lenovo Mirage	Yes	Yes	Yes	Yes	Required
Canon EOS R5 Mark II CRM	Yes	Yes	Yes	Yes	Required
Canon EOS R5 C CRM (8K59.94)	Yes	Yes	Yes	Yes	Required
CALF Gen 1	Yes	Yes	Yes	Yes	Required
Vuze XR VR180 dual-track 5.7K30 / 4K60	Yes	Yes	Yes	Yes	Required
Vuze XR VR180 assembled 4K30	Yes	Yes	Yes	Yes	Required

NOTE

Full-version coverage assumes the camera is included in your edition

The Yes entries describe registered routes in the full version; demo editions can compile out delivery or camera capabilities. Every listed full-version video profile requires calibration, and optional camera families such as CALF Gen 1 still follow the installed edition.

Photo profile coverage

Photo route	Preview/export	Calibration behavior
GoPro MAX2 .36P	JPG export and Timeline	Uses same-serial video calibration when present; otherwise opens with transient photo alignment after a warning.
DJI Osmo 360 JPG	JPG export and Timeline	Uses a transient single-frame photo correction; no video calibration or serial chooser is needed.
Insta360 X5/X6 INSP/DNG	JPG, DNG from DNG sources, and Timeline	Requires matching same-camera-serial video calibration. X6 uses its own lens geometry.
Kandao Q3U JPG/DNG	JPG, eligible DNG, and Timeline	Requires a compatible Q3U video calibration; a chooser may be offered.
Lenovo Mirage JPG	JPG export and Timeline	Requires matching same-serial video calibration.
Canon EOS R5 Mark II JPG/CR3	JPG, processed DNG from CR3, and Timeline	Uses a transient single-frame photo correction; it does not require or create persistent video calibration.
Vuze XR paired JPG/JP2	JPG export and Timeline	Creates its own same-serial photo calibration; no video is needed.
Vuze XR paired DNG/DN2	JPG, processed DNG, and Timeline	Creates its own same-serial RAW-photo calibration; keep both original eyes.
Vuze XR assembled JPG	JPG export and Timeline	Creates its own same-serial assembled-photo calibration; no companion is required.

Vuze XR: VR180 originals

Capture	Files to keep	How to open it
5.7K30 or 4K60 dual-track video	The original MP4, including both video tracks and motion metadata.	Open the MP4 directly. Camera RAW/unstitched video is encoded video, not a Bayer photo.
4K30 assembled video	The original VR180 MP4.	Open the MP4 directly; Studio reads its side-by-side projection and camera motion.
Unstitched JPEG photo	Matching HET_####.JPG and HET_####.JP2 in the same folder.	Select either member; Studio opens one stereo photo. JP2 is this camera's JPEG companion, not JPEG 2000.
RAW photo	Matching HET_####.DNG and HET_####.DN2 in the same folder.	Select either member; Studio develops both sensor images together.
Assembled JPEG photo	The original 6000x3000 JPG.	Open this single file; it already contains both eyes.

NOTE

VR180 scope and platform validation

Only Vuze XR captures recorded in VR180 mode are supported. Vuze 360 recordings and externally converted files are outside this support. As of 6 November 2026, the original-media preview, photo, stabilized-video export, Timeline, and streaming workflows have been verified on macOS. Windows validation with the Vuze camera originals is still pending; check your delivery notes for its validation status.

NOTE

Canon body and lens support is exact

EOS R5 Mark II with RF 5.2mm supports CRM and exact 8192x5464 JPG/CR3 photos. EOS R5 C with that lens supports CRM at 8192x4320 and 60000/1001 fps; R5 C photos and other recording variants have no registered route. An identified but unsupported rig is still rejected.

Canon CRM first-use workflow

1	**Open the original CRM** Studio opens a neutral native GPU preview before full visual motion analysis, so you can inspect and calibrate the source immediately.
2	**Check the integrated RAW look** Studio develops Canon RAW with its built-in recipe and packaged color resources. No separate processing application is needed.
3	**Calibrate video geometry** Complete Auto-Calibration for the matching R5 Mark II or R5 C video profile before final preview, export, Timeline, or streaming.
4	**Enable stabilization when needed** Studio generates and caches the visual trajectory on demand. R5 Mark II combines optical motion with its camera metadata; R5 C uses its separate optical-only trajectory.
5	**Monitor audio** CRM exposes four 48 kHz, 24-bit mono channels. Monitoring defaults to channels 1 and 2.

> **NOTE**
> **Weak row evidence remains usable**
> R5 Mark II measures rolling-shutter phase and readout per CRM. When row evidence is weak, its ordinary trajectory remains available without exact row correction. R5 C has no IMU or row correction in its registered optical-only route.

Companion-file rules

DJI multipart Keep all sibling OSV members. Studio validates identity, order, stream compatibility, duration, and logical revision as one inseparable recording.	**Insta360 split media** Keep matching _00_ and _10_ parts together. Missing or mismatched parts fail before rendering.
Vuze XR photo pairs Keep JPG with JP2, or DNG with DN2, using the same base name in the same folder. Do not rename the secondary file or combine files from different captures.	**Timeline references** Do not move or replace source media after queuing. A changed logical source is rejected instead of silently rendering different content.
GoPro chaptered video Keep every GS<FF><NNNN>.360 member together, beginning with GS01. Multi-member GS02+ playback remains fail-closed until genuine chaptered camera footage completes end-to-end validation.	**Customer editions** A camera can be valid but absent from a particular build. Install the edition made for that camera family.

04

EVERYTHING YOU NEED, WITHOUT A MAZE OF PANELS

Workspace and import

The left column is your loaded-media shelf, the center is the native GPU preview, the bottom is transport and range control, and the right rail opens processing, information, Timeline, and queue views.



The standard video workspace with a calibrated DJI Osmo 360 ride loaded at 8K50.

Import one or many files

1	**Choose media** Use File > Open the files, the Open Files/Add files button, or drag supported files into the window.
2	**Wait for the batch** Studio probes and validates the entire selection before it exposes the first active clip.
3	**Read the summary** If some files are rejected, one summary lists every rejected name and reason. Accepted media arrives together.
4	**Handle calibration** If the active import needs calibration, that flow starts only after loading and the rejection summary are finished.

> **TIP**
> **Fast clip switching**
> While a clip remains in the left column, Studio keeps its parsed metadata, prepared trajectory, decoder, and native preview state warm. Remove the clip only when you are ready to release those caches.

Per-media session state

- Every loaded medium keeps its own color, LUT and strength, anti-flicker, image correction, audio mode, and stabilization choices for the current session.
- Switching media restores that medium's state; it never inherits the settings of the clip that was just visible.
- Restored grade values remain exact, but the named color-profile selector returns to (None) because the session snapshot stores values rather than a live preset association.
- Removing a medium forgets its session state. Remove All clears the whole session, and restarting Studio does not restore these creative settings.
- Aliases of the same logical multipart recording share one state because Studio treats them as one inseparable medium.
- A Timeline clip starts from the source medium's current snapshot, then becomes independently editable inside the project.

Transport controls

Control	Action
Previous Frame	Move back one source frame.
Play / Pause	Start or pause without intentionally dropping late source frames.
Next Frame	Move forward one source frame.
Timecode	Shows current time and clip/range duration.
[In	Set the export/stream start at the current frame.
] Out	Set the export/stream end at the current frame.
Volume	Adjust monitoring volume. It does not replace the chosen audio-processing mode.
Stream	Open DeoVR streaming settings for a video or Timeline.
Export	Open Video Export Settings or Photo Export Settings for the active media.

Playback details that matter

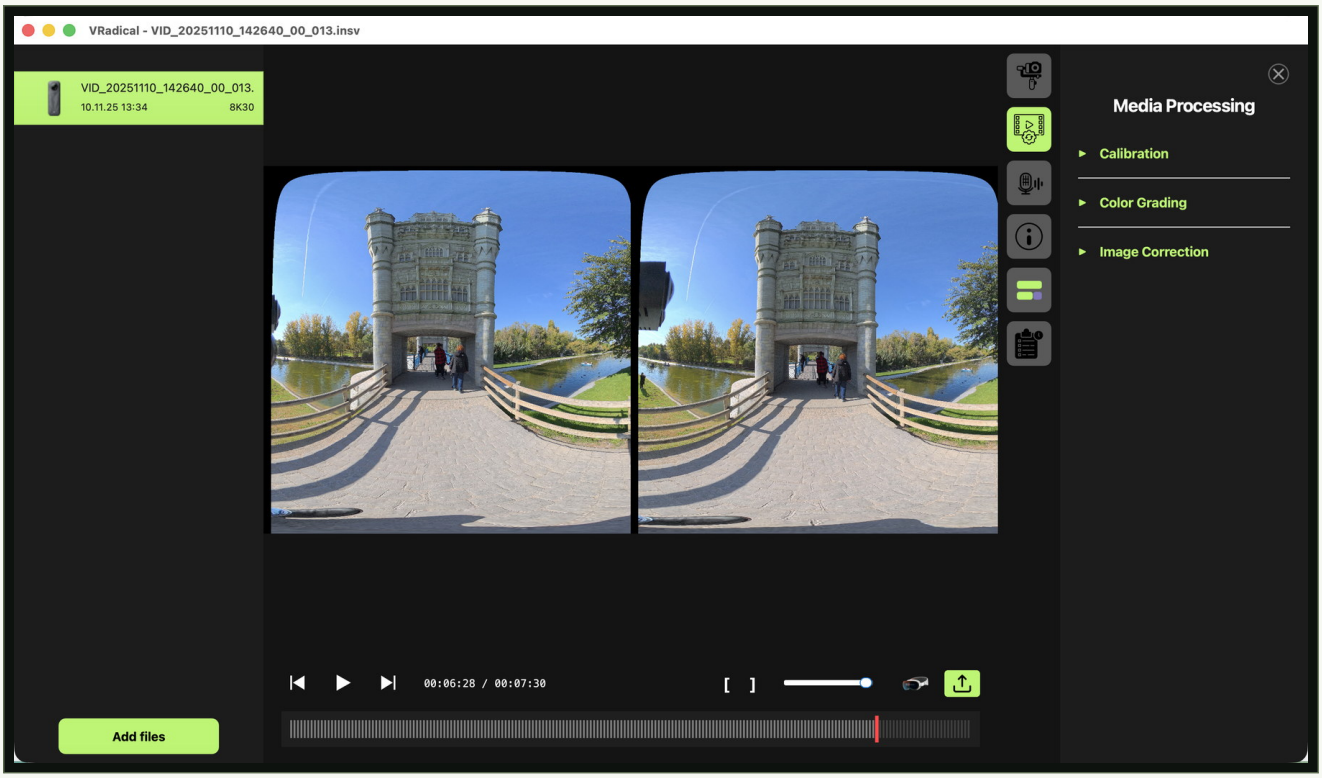
- Fractional rates such as 29.97 and 59.94 use a high-resolution schedule rather than an integer-millisecond timer.
- When processing is late, Studio displays frames as soon as possible instead of skipping source frames to catch up.
- At end of playback, the last rendered frame stays visible. Press Play to restart from the active range.
- The preview remains centered when the window is resized.
- Photos intentionally hide playback, audio, stabilization, streaming, and In/Out controls.

05

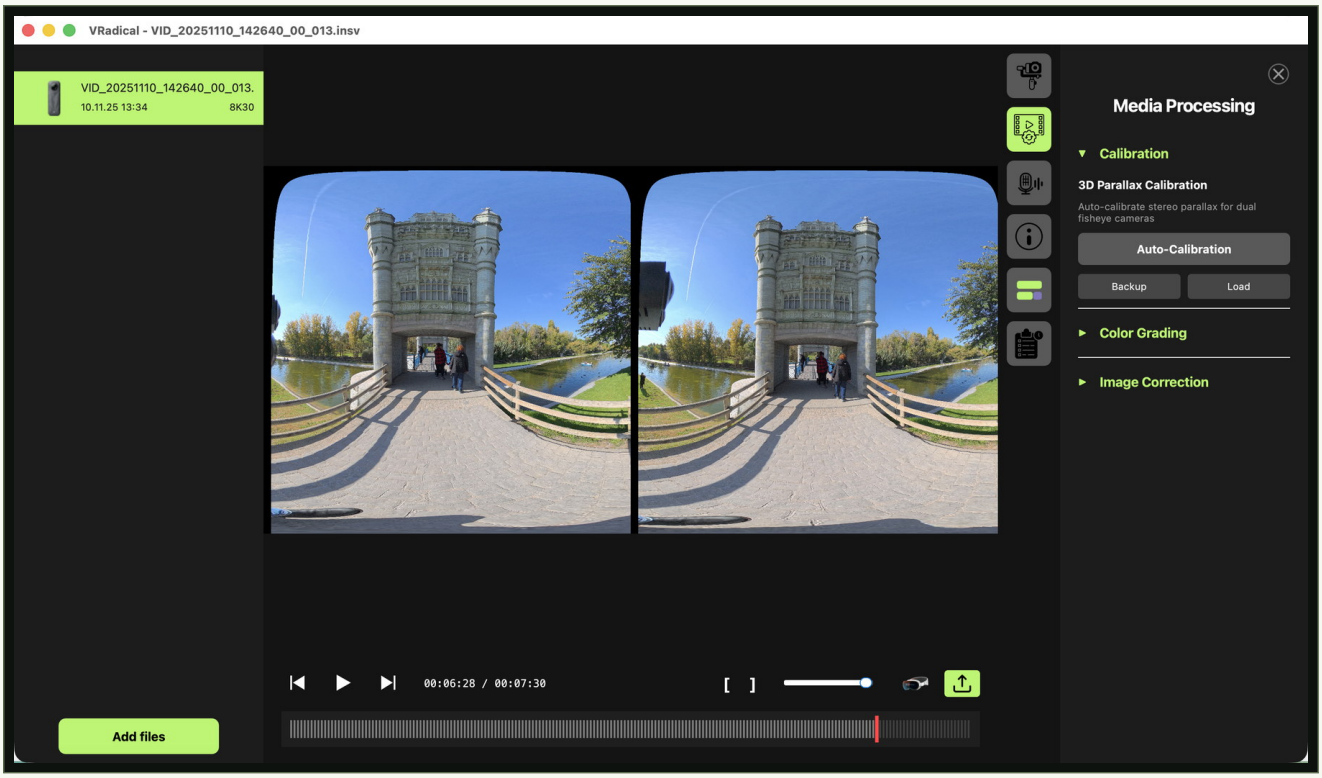
PERFECT PARALLAX IS A SAVED CAMERA ASSET

Calibration

Stereo calibration corrects the final relationship between both output eyes. Studio stores it by camera/profile and automatically reapplies it on future loads. When a profile requires calibration, missing or invalid data blocks final preview, export, streaming, and Timeline output.



Media Processing groups calibration, color grading, and image correction in one focused panel.



The 3D Parallax Calibration controls: Auto-Calibration, Backup, and Load.

Automatic calibration workflow

1	**Load a calibration-capable source** Use a clear, textured scene with useful detail across both eyes. Videos need several useful moments; supported photo calibration uses the still itself.
2	**Open Media Processing** Expand Calibration and choose Auto-Calibration. Confirm replacement if a usable calibration already exists.
3	**Let Studio extract samples** The native GPU renders the final-output eye pairs needed by that source. Do not move, rename, or disconnect the source during the run.
4	**Let the solver finish** Studio prepares stereo matches, rejects weak samples, solves shared rectification and output translation, and validates the result.
5	**Load automatically** The resulting camera/profile INI is saved and applied immediately. The same profile loads automatically next time.

> **NOTE**
> **Safe replacement**
> If a new solve fails, Studio keeps the previous usable calibration. Backup before a deliberate recalibration, and Load only a file whose camera/profile identity matches.

Modified DJI sensor orientation

- On first setup, Studio may determine cardinal sensor orientation before parallax calibration.
- Automatic setup measures stabilization ON and OFF states, then saves one complete Mod profile.
- If the scene is ambiguous, Studio does not guess. Use the calibration-purpose manual preview to capture both states, then retry calibration.
- An existing Mod profile is authoritative, including an all-zero profile created elsewhere.

Photo calibration identity

Situation	What Studio does
A photo route uses matching video calibration	Loads that route's same-serial video calibration automatically.
Vuze XR VR180 photo	Creates a persistent calibration from the photo itself, specific to its serial and paired-JPEG, paired-RAW, or assembled-JPEG profile.
Q3U photo serial differs	Shows a chooser containing compatible, usable Q3U video calibration serials.
Osmo 360 JPG or R5 Mark II JPG/CR3	Runs a transient single-frame correction; no persistent video-calibration INI is required.
A route requiring video calibration has none	Asks you to load and calibrate a compatible video before opening the photo.
Q3U chooser is cancelled	Rejects the photo because the required calibration identity was not selected.
Alias target was removed	Cleans the broken alias and offers the chooser again on the next preflight.

Calibration troubleshooting

Symptom	Best next move
Missing	Run Auto-Calibration with a source supported by that camera profile. Vuze XR photos calibrate from the photo itself.
Invalid	Verify the camera/profile identity, then create a fresh solve. Do not reuse an unrelated INI.
Low-feature failure	Choose a scene with more texture, less motion blur, and clear overlap across both eyes.
Manual DJI setup requested	Capture both stabilization states. A one-state draft is intentionally not saved as authoritative.
Photo has no calibration	Follow the camera's photo policy. Vuze XR photos create their own calibration without a video.
Canon CRM has no calibration	Run Auto-Calibration from the matching R5 Mark II or R5 C CRM; neutral preview may open first, but final video routes require calibration.

Vuze XR calibration is specific to each capture type

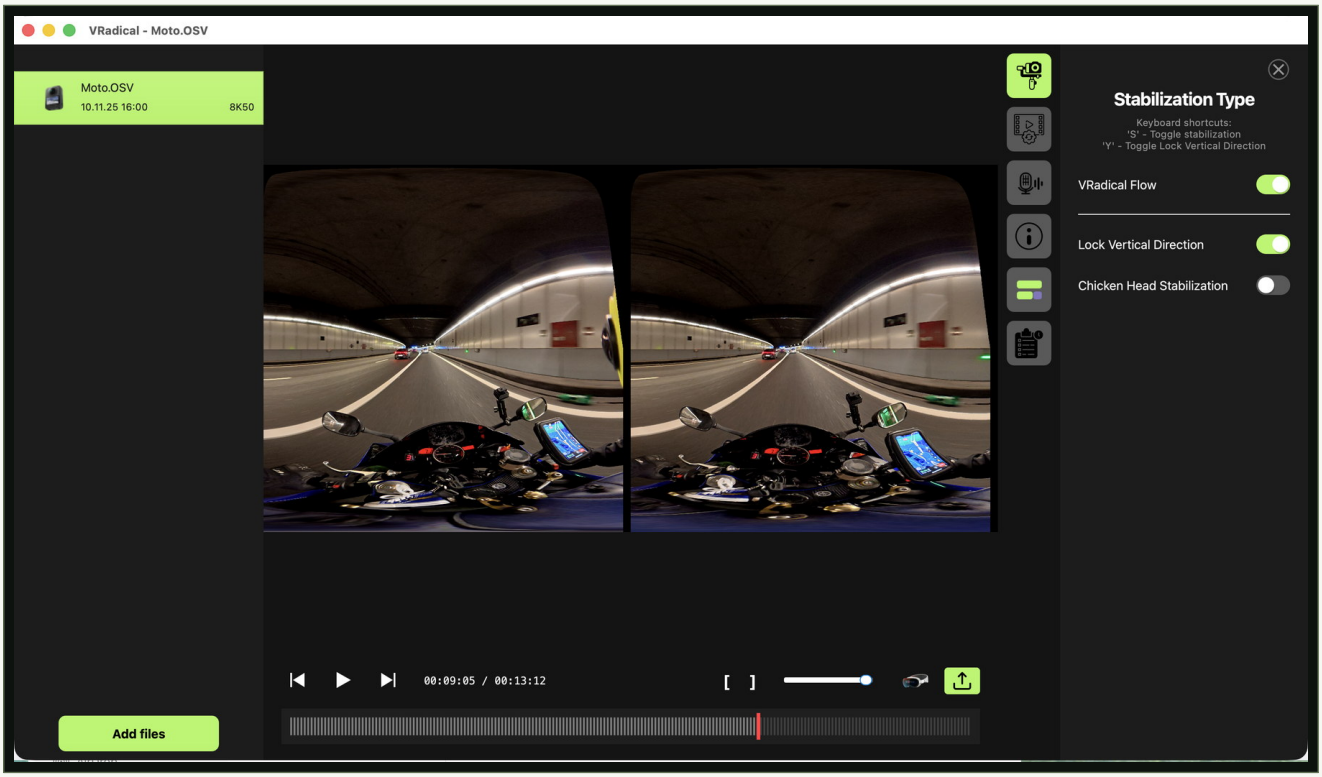
Studio keeps separate calibration for dual-track video, assembled video, paired JPEG photos, paired RAW photos, and assembled JPEG photos. Each belongs to the same physical camera serial and its capture type. Complete the guided calibration when first opening a new type; a calibrated video does not replace photo calibration. For any of the three photo types, Auto-Calibration works from that photo alone and saves a reusable profile.

06

CHOOSE HOW THE VIEWER EXPERIENCES THE MOTION

Stabilization

Stabilization controls are camera-policy aware. Studio only shows controls that make sense for the active profile, while preview, export, streaming, and Timeline use the same prepared motion contract. VRadical Flow is the master switch; vertical lock and Chicken Head are optional modifiers.



DJI Osmo 360 in motion: VRadical Flow and Lock Vertical Direction are enabled while Chicken Head remains off.

Control	Use it when	Viewer experience
VRadical Flow	Most handheld or moving-camera footage.	Master ON/OFF control that smooths global motion while retaining intentional movement.
Lock Vertical Direction	You want a steady horizon/vertical reference.	Flow modifier that holds the horizon/pitch reference. It is not a yaw lock.
Chicken Head Stabilization	You want yaw held to the clip's initial heading.	Flow modifier that holds initial yaw. Enabling it also enables Lock Vertical Direction in the main, export, and streaming dialogs.

A reliable setup order

1 **Turn on VRadical Flow**
Start with the camera policy defaults and play a representative moving section.

2	**Decide on vertical lock** Enable Lock Vertical Direction if horizon stability matters more than retaining the camera's pitch motion.
3	**Try Chicken Head only when intentional** Use it for a fixed-forward experience, not as a generic extra smoothing switch.
4	**Check source limits** Large framing corrections or extreme motion can expose honest black borders where no source pixels exist.
5	**Preview the export range** Scrub the first, fastest, and last motion in the marked range before export.

Camera-specific controls

- GoPro BERS1 uses fixed native rolling-shutter behavior. Gyro-fusion and residual rolling-shutter sliders remain hidden.
- Older GoPro BERS0/EAC profiles can expose Gyroscope and camera-specific rolling-shutter factor controls.
- DJI Osmo 360 and Osmo 360 II use their own camera metadata and rolling-shutter policy, with separate calibration identities and one rigid motion basis for both eyes.
- Insta360 EVO, X4, X5, X6, One R/RS, and 1-inch profiles use VRadical Flow with dense camera motion metadata, automatic metadata-based rolling-shutter correction, and one shared trajectory for both eyes. Residual rolling-shutter sliders stay hidden.
- On X4/X5 clips recorded with both sensors down, Studio can retry left +90/right -90 automatically when no authoritative Mod profile exists, then save the working orientation. X6 keeps its own source layout policy.
- Kandao Q3U video automatically uses VRadical Flow's moving-camera trajectory path shared with Insta360 X4/X5/X6, or its steadier static path when the whole recording is detected at rest. Rolling-shutter correction is automatic with complete metadata. It supports the 24, 25, and 30 fps families within a 0.5 fps tolerance. Inspect tripod shots from start to end for residual drift.
- Canon CRM offers Off, VRadical Flow, Lock Vertical Direction, and Chicken Head. R5 Mark II uses optical motion with its CTMD inertial data and applies exact rolling-shutter correction only when per-clip evidence proves it. R5 C has a distinct optical-only trajectory without IMU or row correction. Gyro-fusion and residual rolling-shutter controls remain hidden.
- Vuze XR VR180 video uses VRadical Flow with the camera's validated motion metadata. Lock Vertical Direction and Chicken Head are available for both dual-track and assembled video. Vuze rolling-shutter correction is not available; a still photo has no temporal stabilization.
- Photos are static sources and do not expose stabilization controls.
- Legacy Follow Me fields are compatibility data, not a supported public mode. Use Lock Vertical Direction and Chicken Head for supported output.

> **NOTE**
> **Rally chooses its own motion contract**
> Selecting Rally output forces VRadical Flow, Lock Vertical Direction, and Chicken Head Stabilization on and disables those controls. Returning to another projection restores the choices you had before Rally.

07

MAKE IT FEEL FINISHED

Look, geometry, and audio

Media Processing is where the technical image becomes your image. Color and LUT work is camera-neutral, image correction changes framing and eye geometry, and audio modes flow into preview, Timeline, and export.

Color grading controls

Control	Practical use
LUT	Choose a built-in camera-aware look or import a 3D .cube file. The LUT runs first in display-referred BT.709 signal space.
Intensity	Blend the LUT from 0 to 100 percent before color sliders.
Temperature / Tint	-100 to +100. Bradford white balance runs in linear light; positive Tint moves toward magenta.
Saturation	-100 to +100. Scales chroma while preserving luminance.
Exposure	-5.0 to +5.0 EV in 0.1 steps. A +1 EV move doubles linear-light exposure.
Contrast	-100 to +100. A luminance-only S-curve preserves in-gamut chromaticity.
Highlights / Midtones / Shadows	-100 to +100. Post-exposure tonal masks change luminance only in the selected zone.
Whites / Blacks	-100 to +100. Set the bright and dark endpoints without per-channel clipping.
Anti-Flickering	Use on video when exposure or lighting flicker is visible. It is not available for photo export.

> TIP
> **Color stays consistent across GPUs**
> Metal and Direct3D 11 use the same ordering and math. Out-of-gamut color is fitted by scaling chroma around luminance, so hue is retained instead of clipping individual channels.

Automatic camera looks and the LOG badge

- Studio reads camera-written color metadata. A LOG badge on the media row and the exact recording format in File Properties tell you when a recognized LOG or flat recording mode is present.
- The Automatic LUT default chooses a validated camera transform when available: for example MAX2 GP-Log, Osmo 360 or 360 II D-Log M, Q3U K-Log, X5 Flat, and X6 I-Log use their matching packaged looks.
- Recognized footage without a validated transform stays neutral; Studio does not invent a LUT. A manual LUT choice, including None, overrides Automatic for that clip and its output.
- The same automatic or overridden look reaches preview, export, Timeline, and DeoVR streaming.

Live RGB histogram

1	**Enable the scope** In Color Grading, select RGB histogram below Reset. The complete final left eye stays visible; the right preview half becomes one graph.
2	**Choose curves** Red, Green, Blue, and Luminance start selected. Turn off unneeded curves while leaving at least one selected.
3	**Compare edits** The graph measures the final graded left eye, including the LUT and visible black borders. Its vertical scale is fixed from the first complete frame so later grade changes remain comparable.
4	**Use Log scale if useful** Log scale changes only graph height, not image processing or the real pixel-count labels. Toggle the scope off and on to establish a new scale.

> **NOTE**
> **A preview tool, not an output effect**
> The graph combines RGB and BT.709 luminance over 256 visible tonal bands. It shows 0-255 for 8-bit sources or 0-1023 for high-precision sources. It never enters video, photo, Timeline, or streaming output and is not saved in grade profiles. The Timeline RGB button beside audio normalization controls the same view.

Import and save a LUT workflow

1	**Open Color Grading** Expand the section and choose New beside LUT.
2	**Select a .cube file** Studio copies the LUT into your personal VRadical LUT folder so it remains available.
3	**Set intensity** Start around 30 to 60 percent, then tune exposure and tonal sliders. A manual choice overrides the camera's Automatic look.
4	**Save a profile** Use Save to name the complete grade for reuse.
5	**Reset safely** Reset returns the active clip's grade to neutral without deleting saved profiles or LUT files.

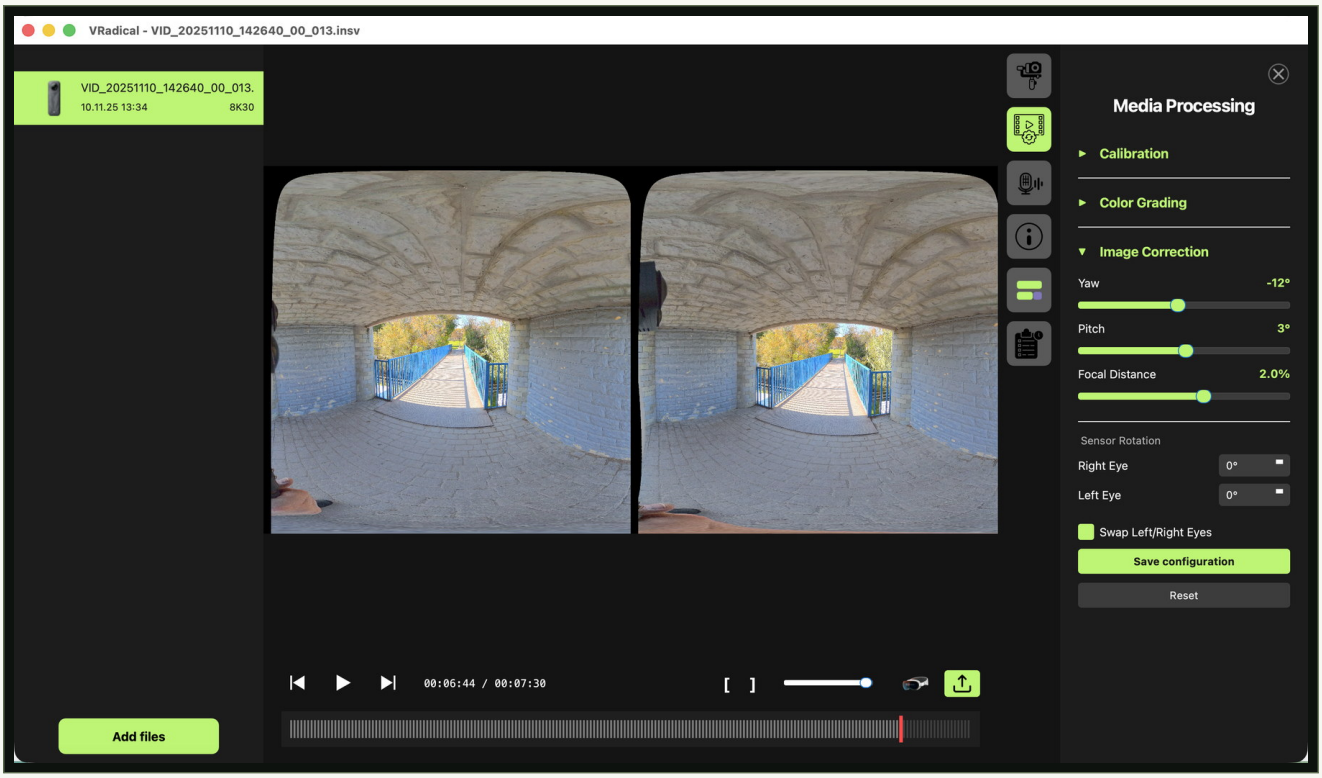


Image Correction for framing, sensor orientation, eye order, and per-camera Mod profile saving.

Image correction controls

Control	Range / behavior
Yaw	-180 to +180 degrees. Reframes horizontally in the camera-appropriate projection stage.
Pitch	-180 to +180 degrees. Reframes vertically; source limits can reveal black regions.
Focal Distance	-15% to +15%. Adjusts framing scale without pretending missing source pixels exist.
Faded edges	GoPro routes only when applicable. Softens eligible outer source edges; width is camera-policy controlled.
Left / Right sensor rotation	Cardinal values -90, 0, 90, or 180 degrees for final output eyes.
Swap Left/Right Eyes	Changes the final SBS eye order before calibration is applied.
Save configuration	Writes the active camera's persistent Mod profile.
Reset	Restores the active clip's image correction defaults.

WARNING

Video sensor geometry is not photo framing

Stitched photos use their own route. Studio does not copy video sensor rotations, swap, or focal distance into an already stitched photo's render settings. Camera-specific photo alignment is measured and sanitized separately.

NOTE

Geometry order protects source detail

GoPro BERS0/BERS1/Fisheye and supported Insta360/Q3U routes apply yaw, pitch, and focal correction in the native source projection, before rolling shutter and final sampling. Eye swap happens before calibration, and calibration plus the common stereo visibility mask stay last.



Audio Processing modes are simple on purpose and carry consistently into the delivery path.

Audio modes

Mode	Result
None	Leaves sample processing bypassed.
Focus Vocal	Prioritizes speech and reduces competing ambience when the audio DSP is available.
Noise Reduction	Reduces steady background noise while preserving a more general mix.
Mute	Outputs silence for the clip. This remains available even if advanced audio DSP is absent.

08

FULL-RESOLUTION DELIVERY, RESPONSIVE GPU PREVIEW

Photo workflows

A photo is not treated as a one-frame video. Studio decodes it once, caches the active native GPU source, shows a bounded interactive preview, and keeps the requested or full resolution for export.

Open and finish a photo

1	**Import the original** Open a supported .36p, .jpg, .cr3, .insp, or .dng whose camera family is enabled. Vuze-enabled editions also accept the paired .jp2 and .dn2 entry points. Keep the matching companion beside the selected file.
2	**Resolve geometry** Follow the active camera's calibration flow. Vuze XR photos create their own same-serial calibration without a video; other cameras follow their individual policy.
3	**Inspect both eyes** Check vertical alignment, scale, seams, and common visibility at the center and borders.
4	**Frame the shot** Use Color Grading and supported image-correction controls. Photos do not expose audio or stabilization.
5	**Export** Choose JPG final SBS/VR180, or DNG only when the source route and output contract support it.

Camera-specific photo behavior

Camera	What to expect
GoPro MAX2 .36P	A baked ERP photo with camera-specific yaw/split alignment and optional same-serial persistent video calibration.
DJI Osmo 360 JPG	Stitched ERP seam recovery and transient single-frame stereo correction; no video calibration or serial chooser is needed. Osmo 360 II has no distinct registered photo route.
Insta360 X5/X6	Native INSP/DNG photo route with matching same-camera-serial video calibration. X6 keeps its own upright SBS lens layout.
Kandao Q3U	Dual-fisheye JPG/DNG route with static gravity redress, shared lens orientation policy, and compatible video calibration.
Lenovo Mirage	Extended-XMP photo route with matching same-serial video calibration.
Canon EOS R5 Mark II	Exact 8192x5464 JPG/CR3 route for the RF 5.2mm lens. JPG can export JPG; CR3 can export JPG or a processed DNG. R5 C photos are not supported.
Vuze XR VR180	Paired JPG/JP2, paired DNG/DN2, and assembled JPG each have their own photo calibration. Photos can enter mixed video/photo Timelines. Direct photo preview does not offer a separate DeoVR stream.

Develop and export a Vuze XR RAW photo

1. **Keep both eyes**
   Place the original DNG and DN2 with the same base name in one folder. Open either member as a single stereo photo.
2. **Complete photo calibration**
   Use the photo's own Auto-Calibration. No calibration clip or another camera serial is needed.
3. **Inspect the developed image**
   Studio develops both sensors on the GPU using their individual black levels and white balance, with the original camera's DNG baseline tone and highlight treatment.
4. **Adjust the finish**
   Inspect bright areas and both eyes, then apply the color, image-correction, denoise, and sharpening settings you need.
5. **Choose the deliverable**
   Export JPG for a finished image or DNG for processed 16-bit linear RGB SBS. Source resolution produces 5760x2880 from the supported paired photos; assembled JPG produces 6000x3000.

> **NOTE**
> **Keep the original DNG and DN2**
> A Vuze DNG export contains the rendered stereo result after development and geometry. It is not the original Bayer sensor pair. Keep both camera originals for future development; JPG sources export JPG, not DNG.

> **NOTE**
> **What Canon processed DNG means**
> Canon CR3 DNG output is an Adobe-readable 16-bit LinearRaw RGB SBS image rendered by the native GPU at final geometry. It is neither a CFA pass-through nor a copy of the proprietary CR3 payload. Canon JPG sources cannot export DNG.

> **NOTE**
> **Preview size versus export size**
> Large stills use an interactive preview bounded to 3840x1920 SBS. Export does not inherit that preview limit; it renders the chosen or full output resolution.

Photos in a Timeline

- A photo becomes a silent static visual clip with a default duration of 2 seconds.
- Trim or stretch the block to change how long the same rendered frame remains on screen.
- Photo clips support placement, transitions, color/LUT, and image correction.
- Photo clips never expose audio or stabilization controls.
- Timeline export renders the calibrated native still at the Timeline cadence for the block's duration.

09

ONE DIALOG, A LOT OF DELIVERY POWER

Video export

Video export snapshots the active media state, validates the camera route and calibration, then queues a job that uses the same native GPU processing contract as preview. For H.264, HEVC, and AV1 hardware requests on Windows, Studio reports an encoder failure instead of silently switching to software.

> **NOTE**
> **Windows hardware encoders follow the GPU**
> Studio selects NVENC on NVIDIA, AMF on AMD, and Intel QSV for H.264, HEVC, and AV1 on Intel. This choice follows the active Direct3D 11 adapter and is identical for every supported camera. An H.264, HEVC, or AV1 hardware request never falls back silently to software; if it fails, Studio reports the hardware encoder error. For Canon HEVC rates above the Main-tier ceiling, Studio selects the matching backend's High tier and a one-second hardware rate-control buffer automatically.

Export a clip

1	**Set In and Out** Mark the range in the player, or leave the full clip active.
2	**Open Export** Choose the green export button at the lower right.
3	**Name the file** Choose a destination and a meaningful filename. Studio adds a numeric suffix when the name already exists.
4	**Choose the picture** Set resolution, one of the supported VR or flat projections, frame rate, codec, bit depth, denoise, and sharpening.
5	**Confirm motion and metadata** Review stabilization, hardware acceleration, and VR metadata.
6	**Start Export** The validated job enters the Export Queue. You can keep working while it runs.

Core export controls

Setting	Choices and guidance
Bitrate	10 to 300 Mbps normally. Canon CRM RAW, or a Timeline containing it, allows H.264, HEVC, and AV1 requests up to 795 Mbps; the exact hardware encoder may reject rates its GPU or driver cannot support. AV1 Main profile has a separate general limit of 800 Mbps. The 120 Mbps default is unchanged. ProRes and MV-HEVC do not expose a bitrate control.
Resolution	The dialog spans width 640-15360 and height 180-8640; the camera, source-eye resolution, projection ratio, and codec narrow valid combinations.
Projection	Hequirect, Fisheye, Rally, 2D Linear, 2D Wide, 2D Ultra Wide, or 3D Flat when the active camera route permits them.
Frame rate	Source rate or 24, 25, 29.97, 30, 50, 59.94, 60, 90, 100, or 120 fps. A target above the source automatically selects GPU RIFE interpolation.
Encoding	H.264, HEVC, AV1, ProRes Standard, ProRes HQ, plus macOS Vision Pro formats when available.
Color depth	8 or 10 bits when supported by the selected codec and route.
Denoise	The Enable checkbox uses New Style; unchecked is Off. Older saved enabled profiles still submit New Style through this dialog. The CLI separately accepts explicit old mode.
Sharpening	None, Light, Medium, or Strong. Registered Canon CRM video uses its directional RAW-detail model; other sources use adaptive output sharpening. With New Style denoise and native H.264/HEVC/AV1 encoding, the selected model runs once after denoise.
Hardware Acceleration	On by default. On Windows, H.264, HEVC, and AV1 strictly use the encoder attached to the active GPU. If it fails, Studio reports the hardware error; turn this setting off explicitly only when you want software encoding. Image processing remains on the GPU.
VR / 3D metadata	Standard immersive output can include VR metadata. Flat 2D has no VR/stereo metadata; 3D Flat can add conventional side-by-side 3D metadata without marking the picture as spherical.

Choose the output projection

Choice	Picture	Delivery notes
Hequirect	Stereoscopic half-equirectangular VR180, left and right SBS.	The usual immersive master; VR metadata is available.
Fisheye	Stereoscopic fisheye SBS.	Use when the destination expects fisheye instead of a hequirectangular remap.
Rally	Square stereo equirectangular frame with top/bottom eyes.	Selecting Rally fixes Flow, vertical lock, and Chicken Head on.
2D Linear	One rectilinear 16:9 view from the final left eye, 110-degree diagonal field of view.	No VR or stereo metadata.
2D Wide / Ultra Wide	One rectilinear 16:9 view, 125- or 135-degree diagonal field of view.	More coverage with stronger edge stretching; no VR or stereo metadata.
3D Flat	Two Linear 16:9 views side by side in one 32:9 frame.	Optional 3D SBS metadata; never marked spherical.

> **NOTE**
> **Flat resolution follows the source eye**
> Selecting Flat proposes the largest linked 16:9 or 32:9 size that fits one decoded source eye. Studio keeps square pixels and caps later edits to that source bound. For a 3840x3840 source eye, the largest Flat 2D frame is 3840x2160 and 3D Flat is 7680x2160. Flat cannot use Spatial/Immersive Vision Pro codecs or VR publishing.

Raise frame rate with RIFE

1. **Choose a higher target**
   For a 24, 25, or 30 fps source, choose a higher export rate such as 60 fps. The RIFE badge appears in the dialog and queue.
2. **Review the range**
   GPU RIFE creates intermediate frames after camera rendering and New Style denoise, then the result is encoded once. It can require substantial GPU memory and time at high resolution.
3. **Keep edits independent**
   In a mixed Timeline, each source clip is checked independently; interpolation never crosses a cut or gap. Photos hold their image and are not interpolated.

> **NOTE**
> **RIFE compatibility**
> Source rate or a lower target uses ordinary frame scheduling. RIFE cannot combine with the CLI's explicit legacy old denoise mode. No extra RIFE switch is needed in Studio or the standard CLI.

Codec and container guide

Codec	Container	Best for	Notes
H.264	MP4	Broad compatibility	8-bit oriented and less efficient than HEVC at very high VR resolutions.
HEVC	MP4	Modern VR delivery	Strong default for 6K/8K, with 10-bit support when available.
AV1	MP4	Efficient modern delivery	Encoder availability and speed depend on hardware and platform.
ProRes Standard	MOV	Editing masters	Large, robust intermediate for downstream post-production.
ProRes HQ	MOV	High-quality masters	Larger than ProRes Standard.
Spatial Vision Pro	MOV	Apple spatial presentation	macOS-only, 10-bit, and Hequirect.
Immersive Vision Pro	MOV	Apple immersive presentation	macOS-only, 10-bit, and Hequirect.

Built-in YouTube profiles

Profile	Exact built-in settings
Youtube 8K 8bit	8640x4320, 29.97 fps, 200 Mbps, HEVC, 8-bit, Hequirect, hardware acceleration ON, denoise Off, sharpening None, Flow ON, Lock Vertical ON, Chicken Head OFF, VR metadata ON, YouTube ON.
Youtube 8K 10bit	Same settings, with 10-bit color depth.

TIP

Save any other delivery as a profile

Choose a compatible custom combination in the export dialog and use Save. Fixed camera policy remains authoritative, so a profile cannot force an unsupported projection, depth, codec, or stabilization choice.

TIP

Quality check before a long render

Preview the first frame, a fast-motion section, a seam-heavy section, and the final frame. Then make a short marked-range export before committing to a long 8K master.

10

SINGLE SHOTS OR A WHOLE MIXED DELIVERY

Photo, batch, and queue

Photo export keeps the still-image route honest, while mixed batch export validates every selected medium before anything enters the queue. Each camera family and media kind gets the settings it actually supports.

Photo export settings

Setting	Behavior
File name and path	Choose the output folder and final name. JPG is the normal delivery format.
Format	JPG final SBS/VR180, or DNG only for eligible DNG source routes.
Projection	Hequirect SBS/VR180 or Fisheye SBS when the photo pipeline supports it.
Resolution	Use the full/requested still resolution, independent of the bounded interactive preview.
Denoise / Sharpen	Check Denoise: Enable for New Style or leave it unchecked for Off; choose None, Light, Medium, or Strong sharpening.
VR/photo metadata	Enabled for JPG when you want compatible viewers to identify the result.

WARNING

DNG is conditional

DNG output is route-specific. Eligible Insta360/Kandao routes follow their own native contract. Canon CR3 produces an Adobe-readable 16-bit LinearRaw RGB SBS result. Vuze XR DNG/DN2 likewise exports processed 16-bit linear RGB SBS after development and geometry. These are processed images, not CFA or proprietary-payload pass-through. JPG sources cannot export DNG.

Mixed batch export

1. **Select several media items**
   Include supported videos and photos from one or more enabled camera families.
2. **Choose Export**
   Studio groups the selection by camera family and media kind.
3. **Review every tab**
   Each group keeps its original video or photo export controls. A fixed camera policy can override an incompatible tab choice.
4. **Choose the destination**
   Video groups may offer immersive or Flat projections according to every source eye in that group. A higher chosen output rate activates RIFE for eligible video sources.

5	**Resolve names** Choose output names and let Studio avoid collisions with unique suffixes.
6	**Validate the complete batch** Studio checks every photo/video job before queuing any of them.
7	**Enqueue atomically** If validation succeeds, the full batch enters the queue and executes sequentially.

> **NOTE**
> **Every item keeps its own state**
> Batch assembly freezes the media's calibration, Mod profile, stabilization, color, LUT, image correction, and audio state. A non-active clip never borrows settings from the currently displayed clip. Untouched group-level stabilization controls preserve each medium's own state; an explicit change in a batch tab overrides that group.

Export Queue

State	Meaning
Pending	Validated and waiting for earlier jobs.
Starting	Preparing the native route, decoder, processing snapshot, and encoder.
Exporting	Rendering with live percentage progress.
Completed	Delivery finished successfully.
Failed	The job stopped with a preserved error message and can be inspected or retried.
Cancelled	The user requested a clean stop.

Queue context actions

- Export Settings - review the frozen job settings.
- Show in Finder / Explorer - reveal a completed output.
- Retry - rebuild a failed or cancelled attempt from its job contract.
- Show Error - read the complete failure reason.
- Remove from list - clear a finished entry without deleting its output.
- Cancel export - request a clean stop for an active job.

Publish after export

Destination	Workflow
DeoVR	Connect an account, prepare title/description and destination options, then upload after export succeeds.
JillVR	Connect and configure the destination. Upload begins only after the local file completes.
YouTube	Choose title, description, category, privacy, schedule, thumbnail, subtitles, language, and applicable video options.

TIP

Export and upload are separate jobs

A local export must complete successfully before publishing starts. The queue shows upload progress separately, so a network retry never requires re-rendering a good local master.

11

MIX CAMERAS WITHOUT FLATTENING THEIR IDENTITY

Timeline

Timeline mode gives you one visual track for supported video and photo clips plus one external audio track. Every visual clip keeps its named camera pipeline, calibration, and per-clip processing settings.

Build a project

1	**Load source media** Import every video and photo you plan to use. Referenced media also loads when you open a Timeline project.
2	**Enter Timeline** Choose the Timeline button on the right rail.
3	**Add visual clips** Double-click or drag supported media into the visual track.
4	**Add external audio** Use + Audio for WAV, MP3, M4A, AAC, or FLAC from the file picker.
5	**Arrange and trim** Drag clips, adjust edges, split at the playhead, and remove unwanted blocks.
6	**Polish each clip** Open clip-specific audio, color, image correction, or stabilization where supported.
7	**Save** Write a .vradical-timeline project before export or streaming.

Editing operations

Operation	Behavior
Move	Reposition a selected visual or audio block on its track.
Trim	Drag a block edge to change source In/Out for video or displayed duration for a photo.
Split	Cut the selected video at the playhead with the toolbar or C/B shortcut.
Remove	Delete the selected block without deleting the original media.
Mute	Mute a selected video clip's embedded audio.
Zoom / Fit	Change Timeline scale or fit the project span into view.
Normalize audio	Analyze levels and apply a consistent Timeline listening target.
RGB	Toggle the same final-left-eye RGB/luminance histogram used in Color Grading. The button sits next to audio normalization.
Track volume	Adjust external audio-track level independently.

Transitions

Track	Choices
Video	None, Fade to Black, Fade to White.
Audio	Fade, Crossfade, Keep as is.

Per-clip settings

- Video clips can carry their own stabilization mode, audio mode, color/LUT grade, and image correction.
- A new clip is initialized from its source medium's current session snapshot; later Timeline edits belong to that clip.
- Photo clips carry color/LUT, image correction, placement, trim duration, and transitions, but no audio or stabilization.
- Sensor rotations and Mod profiles are materialized per camera serial before a job is assembled.
- A Timeline may mix clips from different registered camera profiles on the same visual track.
- If a referenced logical multipart source changes, Studio fails closed and asks you to reload instead of rendering new physical members silently.

Save, reopen, export, and stream

- Save projects with the .vradical-timeline extension.
- Keep project media at stable paths, with companion and multipart files intact.
- Opening a project imports referenced photos and videos into the left media list.
- Timeline export uses the same camera route per clip and the selected global output cadence/codec.
- A global output cadence above a video's own source rate selects RIFE for that clip. The interpolation pair is reset at every edit boundary; photo clips are not interpolated.

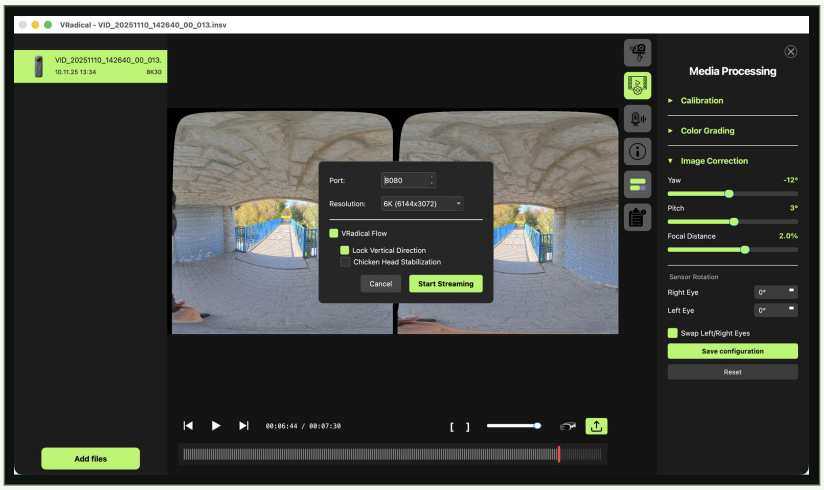
- Flat 2D and 3D output are available only when every contributing camera route and source-eye size meet the selected projection contract.
- Timeline DeoVR streaming uses the same local HLS workflow as a video clip.
- Preview the frames around every cut and transition before a long master export.

12

A LOCAL HEADSET-READY HLS STREAM

DeoVR streaming

Studio can stream a supported video or Timeline over your local network. It prepares hardware-encoded HEVC HLS, publishes a DeoVR entry point, and keeps temporary segments managed for the life of the session.



DeoVR setup on port 8080 with 6K output, VRadical Flow, and Lock Vertical Direction.

Start a stream

1	**Load a video or Timeline** Photos cannot be streamed.
2	**Set the range** Optional In and Out points define the streamed section.
3	**Open Stream to DeoVR** Choose the headset button in the lower-right transport area.
4	**Choose port and resolution** Port 8080 and 4K are the defaults. Available sizes are 4K, 5.7K, 6K, and 8K. Resolutions above 4K require a very powerful GPU.
5	**Review stabilization** Use the same motion choices you previewed for the clip.
6	**Start Streaming** Wait through Buffering until Studio reports Streaming and displays the URL.
7	**Open DeoVR** On a reachable device, open http://<computer-ip>:<port>/deovr.

8 **Stop when finished**
Choose Stop Streaming so the server closes and temporary segments are removed.

WARNING
Above 4K needs substantial GPU headroom
Resolutions above 4K require a very powerful GPU. If rendering cannot keep pace with real-time playback, DeoVR may buffer and Studio may report a playback or streaming error. Test at 4K first, then increase the resolution only after playback is stable.

Video format HEVC HLS with hardware encoding and 2-second segments.	**Ready threshold** Studio normally waits for three playable segments before publishing the stream.
Playlist The media playlist is available at /stream.m3u8 behind the DeoVR entry point.	**Temporary data** Managed in the system temporary folder under vradical_hls and cleaned before, after, and on failure.

WARNING
Use a trusted private network
The local server binds network interfaces without account authentication or TLS and permits cross-origin playback. Use it only on a network you trust, avoid public Wi-Fi, and stop the stream promptly.

Streaming fixes

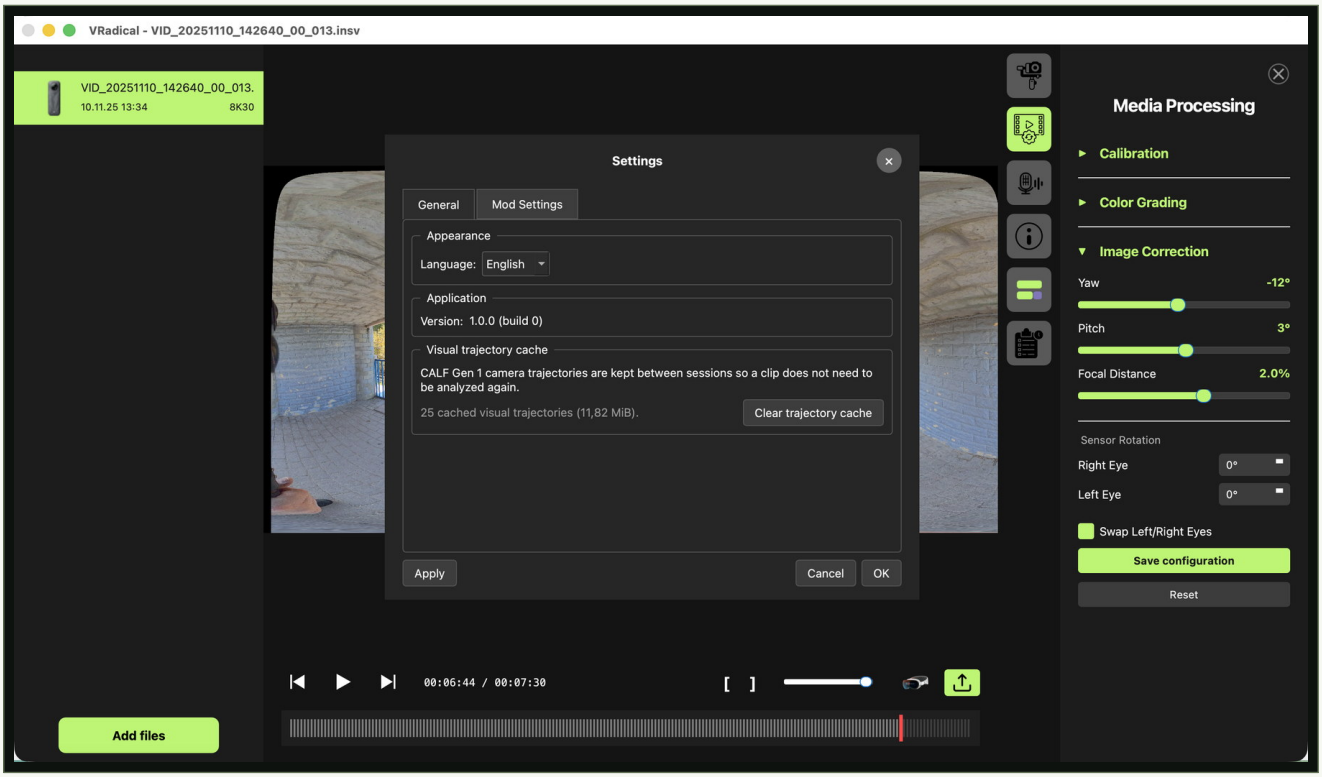
Problem	Fix
Port already in use	Choose another port between 1024 and 65535.
Headset cannot connect	Confirm both devices share a reachable private network and allow the chosen port through the firewall.
Buffering never becomes Streaming	Check hardware encoder availability, free disk space for temporary segments, and the active export range.
URL uses the wrong interface	Use the computer's reachable LAN address with the chosen port and /deovr path.
Photo selected	Load a supported video or Timeline. Static photos do not expose streaming.

13

KEEP RECURRING SETUP OUT OF THE WAY

Settings, updates, and profiles

The shared frameless Settings window exposes version information, language selection, trajectory-cache maintenance, authenticated updates in customer builds, and the per-camera Mod profile editor. Use profiles for physical rig setup, not as a substitute for a clip's creative grade.



Shared VRadical Settings chrome with interface language, version/build, and visual-trajectory cache maintenance.

General

Setting	Use
Open Settings	macOS: VRadical Studio > Preferences... or Cmd+,. Windows: File > Preferences....
Language	English is the current reference interface. Use localized variants only when supplied by the application-wide translation system.
Version / build	Use this when checking delivery notes or requesting support.
Visual trajectory cache	Shows reusable analyzed camera trajectories and their size.
Clear trajectory cache	Removes reusable analysis so affected clips will be prepared again next time.

Authenticated updates

1. **Open Updates**
   In an updater-enabled customer build, choose Help > Check for Updates... or Settings > General > Updates.
2. **Connect GitHub**
   Choose Connect. In Connect VRadical Studio to GitHub, use Copy Code and Open GitHub, complete the device flow in your browser, then return to Studio.
3. **Choose automatic checks**
   Automatically check for updates is off by default. Enabling it shows a privacy consent before Studio contacts GitHub in the background.
4. **Check**
   Use Check Now or Retry. The panel shows GitHub status and Last check without blocking the rest of Studio.
5. **Download**
   Choose Download Update. Downloads are always user-initiated and the signed manifest is verified before installation is offered.
6. **Install safely**
   Choose Install Update.... Studio waits for preview/export/streaming/calibration work to reach a safe state, then installs and relaunches.

NOTE

Automatic-check consent

Studio asks: Allow VRadical Studio to contact GitHub automatically to check this private distribution for updates? No camera, media, project, or customer data is sent.

Update control	Meaning
Connect / Reconnect / Disconnect	Manage the GitHub device authorization used only for this private distribution.
Check Now / Retry	Request the current release status immediately.
Download Update	Fetch the verified installer without installing it yet.
Install Update...	Hand off to the platform installer when Studio is safe to close.
Working... / Cancel	Shows an active update operation or requests a clean cancellation.

WARNING

Platform installation prompts are expected

Windows can show an unknown-publisher UAC prompt before Studio quits, installs, and reopens. macOS uses the embedded signed update services and relaunches Studio. Never run an installer whose source or manifest verification is unexpected.

TIP

Optional GitHub release notifications

On the private distribution repository, choose Watch > Custom > Releases to receive release notifications without subscribing to every repository event.



The Mod Settings editor with a clean profile list and separate stabilization ON/OFF sensor orientation.

Create or edit a Mod profile

1	**Open Mod Settings** Open Settings, then choose Mod Settings.
2	**Choose New or a profile** Create a profile or select an existing camera serial from the list.
3	**Enter the exact serial** Camera serial is a semantic identity. Preserve all characters and boundary whitespace exactly as supplied by Studio.
4	**Set both stabilization states** Choose left/right cardinal sensor rotations for Stabilization ON and OFF.
5	**Set eye order and focal distance** Use Swap Left/Right Eyes and the profile focal-distance default only when the physical rig requires them.
6	**Save and Apply** Save Profile, then Apply/OK. A matching loaded camera updates immediately.

> **WARNING**
> **Do not guess a physical profile**
> Non-zero rotations are not proof that a setup is correct. If first-time automatic orientation is ambiguous, complete the guided manual ON/OFF capture instead of saving an improvised profile.

> **NOTE**
> **Flow changes can select another Mod branch**
> A Mod profile stores separate sensor rotations for stabilization ON and OFF. When VRadical Flow changes, Studio rematerializes the matching branch automatically.

Where Studio keeps personal data

Data	Location / behavior
Calibration and Mod profiles	~/.vradical on macOS; the equivalent user profile location on Windows.
Imported LUTs	~/.vradical/luts.
Calibration scratch	~/.vradical/autocalibrate. Inactive sessions are purged; this is not a result cache.
Visual trajectories	Qt generic data location under VRadical/visual-trajectories/v2.
GUI and export profiles	Native Qt application settings for the current user.
DeoVR/JillVR passwords	macOS Keychain or Windows Credential Manager when used by those login flows.
GitHub updater credential	macOS Keychain or Windows Credential Manager. Tokens and account data are never written to Studio logs.

> **NOTE**
> **Uninstalling is not erasing**
> Replacing or uninstalling the app does not necessarily remove calibration, Mod profiles, LUTs, application settings, caches, or operating-system credential entries.

14

THE SAME PRODUCT RULES, READY FOR REPEATABLE JOBS

Command line

The bundled vradical-cli uses the same shared settings models and camera registry as Studio. It is useful for inspection, scripted exports, Timeline rendering, photo output, and local DeoVR streaming.

Packaged executable

Platform	Path
macOS	/Applications/VRadical Studio.app/Contents/MacOS/vradical-cli
Windows	C:\Program Files\VRadical Studio\vradical-cli.exe

Inspect a camera file

```
vradical-cli --json "/path/to/clip.360"
```

Export an 8K HEVC video

```
vradical-cli "/path/to/clip.360" \
  --export "/path/to/output.mp4" \
  --resolution 8K --codec hevc --depth 10 \
  --bitrate 120M --fps 0
```

Export a Timeline master

```
vradical-cli "/path/to/project.vradical-timeline" \
  --export "/path/to/timeline.mov" \
  --codec prores --resolution 7680x3840
```

Export a flat 2D video

```
vradical-cli "/path/to/clip.insv" \
  --export "/path/to/flat-video.mp4" \
  --output-projection flat-2d --resolution 3840x2160 \
  --codec hevc --fps 0
```

> **NOTE**
> **Exact flat dimensions and automatic RIFE**
> Flat CLI output requires an explicit numeric 16:9 resolution for flat-2d, flat-2d-wide, or flat-2d-ultra-wide, or 32:9 for flat-3d. It must fit the source-eye bound. The shorthand --resolution 8K means 7680x3840, while the built-in Youtube 8K profile uses 8640x4320. --fps 0 keeps the source rate; a higher value such as --fps 60 on 30 fps footage automatically enables GPU RIFE.

Export a supported DNG photo

```
vradical-cli "/path/to/photo.dng" \
  --export "/path/to/photo-output.dng" \
  --photo-format dng --resolution source
```

Start a local DeoVR stream

```
vradical-cli "/path/to/clip.360" \
  --deovr --port 8080 --resolution 4K
```

Export Vuze XR VR180 originals

```
vradical-cli "/media/Vuze/HET_0003.MP4" \
  --export "/exports/vuze-video.mp4" \
  --codec hevc --resolution source --fps 0

vradical-cli "/media/Vuze/HET_0004.DNG" \
  --export "/exports/vuze-photo.jpg" \
  --photo-format jpg --resolution source
```

Use the CLI bundled with a Vuze-enabled edition. The DNG example requires HET_0004.DN2 in the same source folder. To export the processed stereo DNG, choose a new .dng output path and --photo-format dng. For video, --no-stabilization disables Flow; --lockY and --chicken-head-stab select its optional motion controls.

User-facing option groups

Group	Options
Inspect	--json, --dry-run, --build-identity, --describe-color-grading, --describe-image-correction, --describe-audio-processing
Destination	--export PATH, --deovr, --port PORT
Video output	--resolution, --bitrate, --codec, --output-projection hequirect|fisheye|rally|flat-2d|flat-2d-wide|flat-2d-ultra-wide|flat-3d, --depth, --fps
Encoder	--no-hw-accel, --fast-av1, --no-vr180-metadata
Range	--in, --out using microseconds or mm:ss(.sss)
Calibration	--calibrate, --no-auto-calibration, --photo-calibration-serial, --no-mod-profile
Stabilization	--no-stabilization, --lockY / --no-lockY, --chicken-head-stab, --gyro-stab when camera policy allows
Image cleanup	--denoise-mode off|old|new, --deflicker, --sharpening None|Light|Medium|Strong
Audio	--audio-processing none|voice-focus|noise-reduction|mute
Color	Temperature, tint, saturation, exposure, contrast, highlights, midtones, shadows, whites, blacks, LUT, LUT strength
Image correction	Yaw, pitch, focal distance, faded edge, width, left/right sensor rotation, and swap eyes
Photo	--photo-format jpg|dng and photo-safe --resolution source
Insta360 1-inch	--one-r-inch-vertical selects the vertical 1-inch profile when applicable

TIP

CLI habits that prevent surprises

Quote every path, use a period as the decimal separator, run --dry-run before a large render, and press Ctrl+C once for a clean cancellation. Video export and DeoVR streaming are mutually exclusive, and photos cannot be streamed.

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CLEAR MESSAGES, PRACTICAL NEXT MOVES

Troubleshooting

Studio fails closed when a result would be misleading: wrong camera route, missing stereo correction, changed multipart identity, or unsupported hardware. Use the exact message and the table below before changing files or settings.

Message or symptom	What to do
Requires an Apple Silicon Mac	Move the job to an Apple Silicon Mac. There is no Intel/CPU preview mode.
VRadical Studio requires a compatible hardware Direct3D 11 GPU. On non-Apple Windows computers, the GPU must be NVIDIA, AMD, or Intel. CPU fallback rendering is disabled.	Install the current hardware-vendor driver, avoid a software adapter or remote fallback, and relaunch. A physical Apple computer running Windows may use its installed hardware GPU.
This file is not supported by this version of VRadical	Install a version that supports the source camera, or use media supported by the installed version. If the file should be supported, contact support with the untouched original and your exact Studio version.
This file is not compatible with this version of VRadical	The optional CALF Gen 1 source does not belong to this delivered edition. Install the matching customer edition.
Camera/media not authorized	Preserve the original and contact support with the camera serial shown by Studio.
Open one VRadical Timeline file at a time	Open a single .vradical-timeline project, then import any extra source media normally.
Calibration missing / invalid	Complete Auto-Calibration for the active camera and capture type. Vuze XR photos use their own still image; a video calibration does not replace it.
Photo has no compatible calibration	Follow the camera's calibration workflow. Vuze XR photos can create their own same-serial profile from the photo itself; keep both members of any paired source.
Vuze XR file is rejected	Use a Vuze-enabled edition and a VR180 original. Keep JPG/JP2 or DNG/DN2 pairs in one folder with the same base name. An assembled JPG needs no companion.
JPG preview fails after replacing the application	Quit Studio completely, install the complete updated application, and reopen it before loading the original again. Do not replace an application bundle while it is running.
Vuze RAW highlights look magenta in an older build	Install the current Vuze-enabled delivery and reopen the original DNG/DN2 pair. Its native RAW development includes the original baseline tone and highlight treatment; no separate pre-conversion is needed.
Sensor orientation is ambiguous	Use a better scene or complete the guided manual ON/OFF orientation capture. Do not save a guessed profile.
Motion / rolling-shutter metadata incomplete	Return to the untouched camera original and its companion parts. The route intentionally refuses an incomplete trajectory.
R5 Mark II exact rolling-shutter proof is not observable or is ambiguous	Continue with the ordinary Canon stabilization trajectory. Studio remembers this valid per-clip result and omits exact row correction; it does not reject the source or disable global stabilization.
Canon packaged color resource is missing or invalid	Repair or reinstall the complete VRadical Studio package, then reopen the CRM/CR3 source. No separate processing application is required.
Multipart OSV set is ambiguous or changed	Restore every original sibling, name/order, and stream-compatible member; then reload.
Logical media source changed after it was queued	Remove the job, reload the current complete source set, inspect it, and queue again.
Hardware encoder unavailable	Read the reported hardware error first. Try a supported rate, depth, resolution, codec, or current GPU driver. On Windows, turn off Hardware Acceleration only when you deliberately want the slower software encoder; Studio never switches to it silently.
RIFE initialization or memory failure	Try a shorter marked range or lower export resolution on a compatible GPU, then retry. Matching the source frame rate bypasses interpolation.

Message or symptom	What to do
Flat projection or resolution rejected	Confirm the camera route offers Flat and enter a square-pixel 16:9 2D or 32:9 3D size within one source eye. Flat cannot use Vision Pro spatial codecs.
Advanced audio mode unavailable	Use None or Mute, or install an edition built with the audio DSP.
Output path invalid / already exists	Choose a writable folder. Studio can add -01, -02, and later suffixes to avoid collisions.
DeoVR server cannot listen	Choose another port and check local firewall/network rules.
vradical-cli was not found	Reinstall the complete application bundle/installer; do not move the CLI out of its package.
GitHub connection expired or rejected	Open Settings > General > Updates, choose Reconnect, and complete the device flow again. Do not share the device code.
Update download or verification failed	Choose Retry on a trusted network. Do not bypass signature/manifest verification or run a partially downloaded installer.
Install Update is waiting	Finish or cancel active export, streaming, calibration, or other unsafe work; Studio installs only after it reaches a safe state.
Export failed	Open Export Queue, choose Show Error, correct the root cause, then Retry.

Collect useful diagnostics

1	**Record version/build** Open Settings and note the exact version and build.
2	**Record the action** State whether the failure happened during import, preview, calibration, Timeline, export, publish, or streaming.
3	**Copy the exact error** Use Show Error in the queue or copy the complete popup text.
4	**Attach the latest session log** From the newest studio-* folder, include the .log files. On macOS: ~/Library/Application Support/VRadical/logs. On Windows: %LOCALAPPDATA%\VRadical\logs. Studio retains terminal output and a bounded log for each run.
5	**Review before sharing** Logs can contain media/calibration paths, camera serials, and processing settings. Environment variables, OAuth codes, passwords, and tokens are excluded. Redact any path or serial you do not want to share.

Before contacting support

- [ ] Keep the original camera file and all companion/multipart members.
- [ ] Do not transcode the failing source.
- [ ] Note the camera model, exact serial, Studio version/build, operating system, and GPU.
- [ ] Include the exact route: preview, calibration, photo, Timeline, export, publish, or DeoVR.
- [ ] Include the exact error and the smallest reproducible range.
- [ ] Contact ChelouteVR via reddit u/cheloutevr or email cheloute.vr@gmail.com when the installed build instructs you to do so.

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THE PAGE YOU WILL COME BACK TO

Shortcuts and reference

Use these verified shortcuts and checklists to move quickly. Shortcuts are context-sensitive: Timeline editing keys work while the Timeline editor has focus.

Global shortcuts

macOS	Windows	Action
Cmd+O	Ctrl+O	Open supported files
Cmd+,	File > Preferences...	Open Settings
Cmd+Q	Ctrl+Q	Quit Studio

Timeline shortcuts

Shortcut	Action
Esc	Leave Cut mode
Cmd/Ctrl+Z	Undo
Cmd/Ctrl+Shift+Z or Cmd/Ctrl+Y	Redo
C or B	Split selected video at playhead
Delete or Backspace	Remove selection
M	Mute selected video
+ or = / -	Zoom in / out
Left / Right	Move one frame
Shift+Left / Shift+Right	Move ten frames
Cmd/Ctrl+Left / Cmd/Ctrl+Right	Move one second
Up / Down	Previous / next edit point
Home / End	Timeline start / end
J / K / L	Reverse shuttle / stop / forward shuttle
I / O	Mark In / Out
Alt+I / Alt+O	Clear In / Out

Final export checklist

- [ ] The correct camera/profile appears in File Properties.
- [ ] Calibration is loaded and both eyes align in near, mid, and far regions.
- [ ] Sensor rotations and eye order match the final SBS viewer convention.
- [ ] Stabilization behaves correctly at the start, fastest motion, and end.
- [ ] Yaw, pitch, and focal distance do not reveal unintended source borders.
- [ ] LUT and tonal controls look consistent in both eyes.
- [ ] Audio mode and monitoring volume are intentional.
- [ ] In/Out points or Timeline duration are correct.
- [ ] Projection, aspect ratio, resolution, frame rate, codec, depth, bitrate, and metadata match the destination. Check whether a higher frame rate has selected RIFE.
- [ ] A short sample export has been checked in the target player/headset.
- [ ] The final output folder has enough space and the queue is monitored through completion.

Fast glossary

Term	Definition
CFA	Color filter array data in some RAW photo sources. Processed SBS DNG output is rendered linear RGB, not a CFA copy.
D3D11	Direct3D 11, the required native Windows GPU backend.
EAC	Equi-angular cubemap camera layout used by supported GoPro profiles.
HLS	HTTP Live Streaming, used for the local DeoVR stream.
LUT	A 3D color lookup table stored as a .cube file.
Mod profile	Persistent physical camera setup for sensor rotations, eye order, and focal-distance defaults.
P010	10-bit 4:2:0 YUV GPU format used by high-depth native paths.
RIFE	GPU frame interpolation automatically used when the selected video export rate exceeds a source clip's rate.
Flat 2D / 3D	Rectilinear video exports: a single 16:9 eye or two 16:9 eyes in a 32:9 SBS frame.
SBS	Side-by-side stereo output, left eye followed by right eye.
VR180	A stereoscopic 180-degree immersive view.

TIP

You are ready

Start with a short calibrated clip, make one intentional change at a time, and check the result in the final viewer. Once that loop feels natural, the same workflow scales cleanly to photos, mixed-camera Timelines, batch jobs, and headset streaming.